Causal Analysis Report

BPI Challenge 2017 - Offer log.xes.gz-im

Decision points:	128985
Choice sets:	3
Transitions fitted:	5
Significant (p<0.05):	0
Generated:	2026-06-21 15:49

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

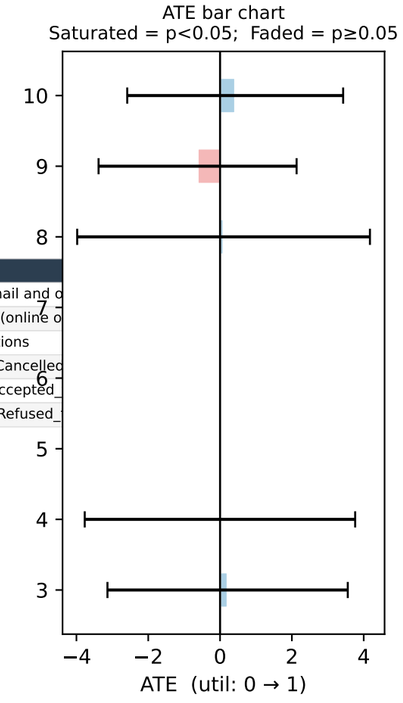
(no data)

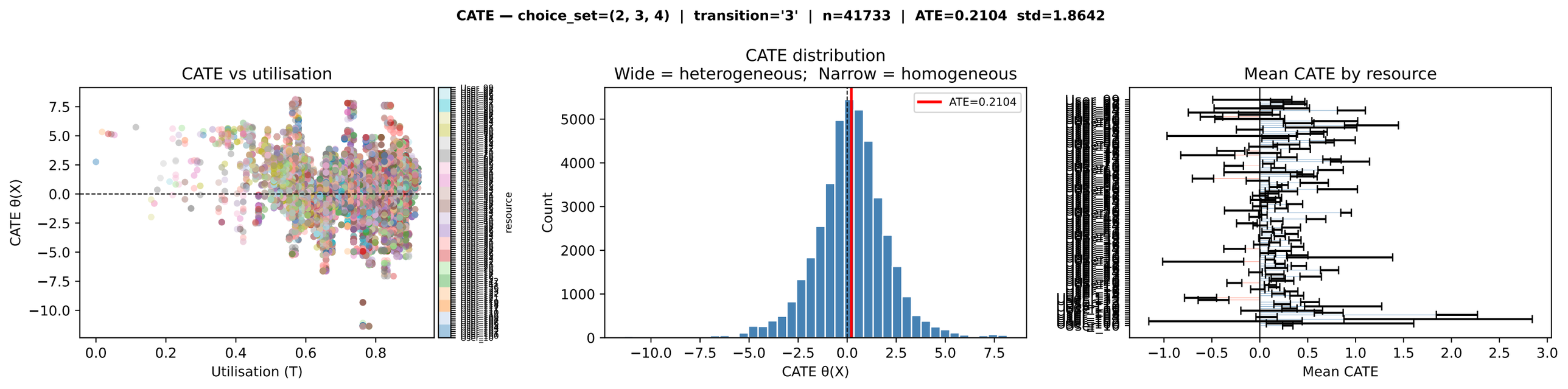
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

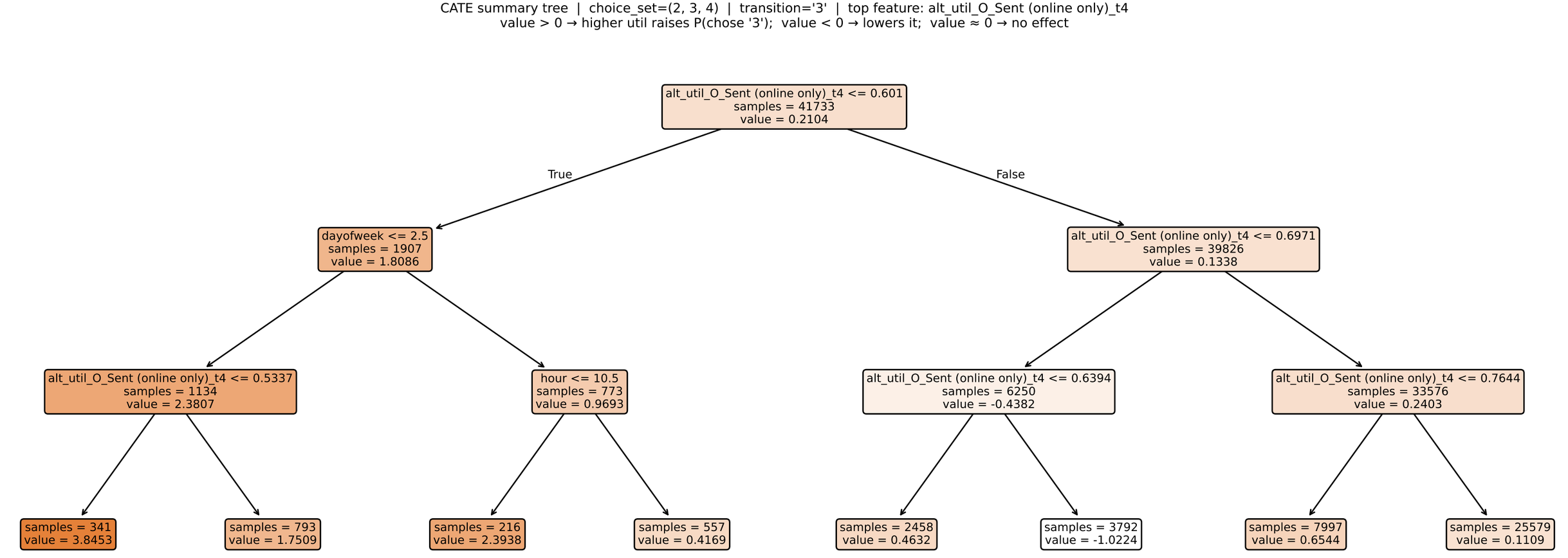
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	3	O_Sent (mail and online)_t3	O_Sent (online only)_t4	0.237002	0.082025	[0.0762, 0.3978]	0.0039	True	θ=+0.2370: 1-unit ↑ util ↑ P(t3/O_Sent (mail and online)_t3) by 23.7%
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	4	O_Sent (online only)_t4	O_Sent (mail and online)_t3	0.031542	0.091883	[-0.1485, 0.2116]	0.7314	False	θ=+0.0315: 1-unit ↑ util ↑ P(t4/O_Sent (online only)_t4) by 3.2%
(5, 6)	('τ_t5', 'O_Returned_t6')	23305	6	—	—	—	nan	nan	—	nan	—	too few observations
(7, 8, 9, 10)	('τ_t7', 'O_Cancelled_t8', 'O_Accepted_t9', 'O_Refused_t10')	42821	8, 9, 10	8	O_Cancelled_t8	O_Accepted_t9, O_Refused_t10	0.102819	0.097026	[-0.0874, 0.2930]	0.2893	False	θ=+0.1028: 1-unit ↑ util ↑ P(t8/O_Cancelled_t8) by 10.3%
(7, 8, 9, 10)	('τ_t7', 'O_Cancelled_t8', 'O_Accepted_t9', 'O_Refused_t10')	42821	8, 9, 10	9	O_Accepted_t9	O_Cancelled_t8, O_Refused_t10	-0.086324	0.034529	[-0.1540, -0.0186]	0.0124	True	θ=-0.0863: 1-unit ↑ util ↓ P(t9/O_Accepted_t9) by 8.6%
(7, 8, 9, 10)	('τ_t7', 'O_Cancelled_t8', 'O_Accepted_t9', 'O_Refused_t10')	42821	8, 9, 10	10	O_Refused_t10	O_Accepted_t9, O_Cancelled_t8	0.366393	0.076225	[0.2170, 0.5158]	0.0	True	θ=+0.3664: 1-unit ↑ util ↑ P(t10/O_Refused_t10) by 36.6%

Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

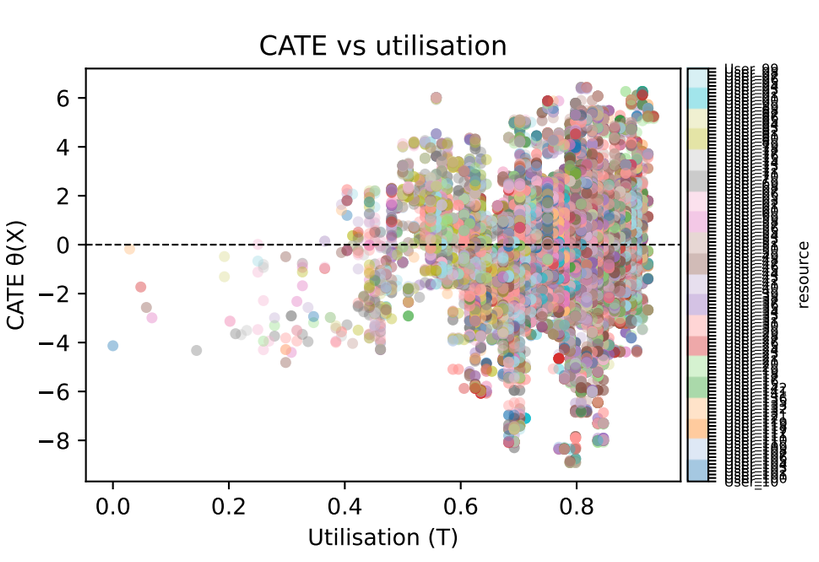
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	3	O_Sent (mail and online)_t3	O_Sent (online only)_t4	0.210368	[-3.1360, 3.5568]	0.9019	False	ATE=+0.2104: 0→1 util ↑ P(t3/O_Sent (mail and o
(2, 3, 4)	('τ_t2', 'O_Sent (mail and online)_t3', 'O_Sent (online only)_t4')	41733	3, 4	4	O_Sent (online only)_t4	O_Sent (mail and online)_t3	-0.003188	[-3.7691, 3.7627]	0.9987	False	ATE=-0.0032: 0→1 util ↓ P(t4/O_Sent (online o
(5, 6)	('τ_t5', 'O_Returned_t6')	23305	6	—	—	—	nan	—	nan	—	too few observations
(7, 8, 9, 10)	('τ_t7', 'O_Cancelled_t8', 'O_Accepted_t9', 'O_Refused_t10')	42821	8, 9, 10	8	O_Cancelled_t8	O_Accepted_t9, O_Refused_t10	0.097606	[-3.9788, 4.1740]	0.9626	False	ATE=+0.0976: 0→1 util ↑ P(t8/O_Cancelled
(7, 8, 9, 10)	('τ_t7', 'O_Cancelled_t8', 'O_Accepted_t9', 'O_Refused_t10')	42821	8, 9, 10	9	O_Accepted_t9	O_Cancelled_t8, O_Refused_t10	-0.625986	[-3.3836, 2.1316]	0.6564	False	ATE=-0.6260: 0→1 util ↓ P(t9/O_Accepted_
(7, 8, 9, 10)	('τ_t7', 'O_Cancelled_t8', 'O_Accepted_t9', 'O_Refused_t10')	42821	8, 9, 10	10	O_Refused_t10	O_Accepted_t9, O_Cancelled_t8	0.421562	[-2.5851, 3.4282]	0.7835	False	ATE=+0.4216: 0→1 util ↑ P(t10/O_Refused_

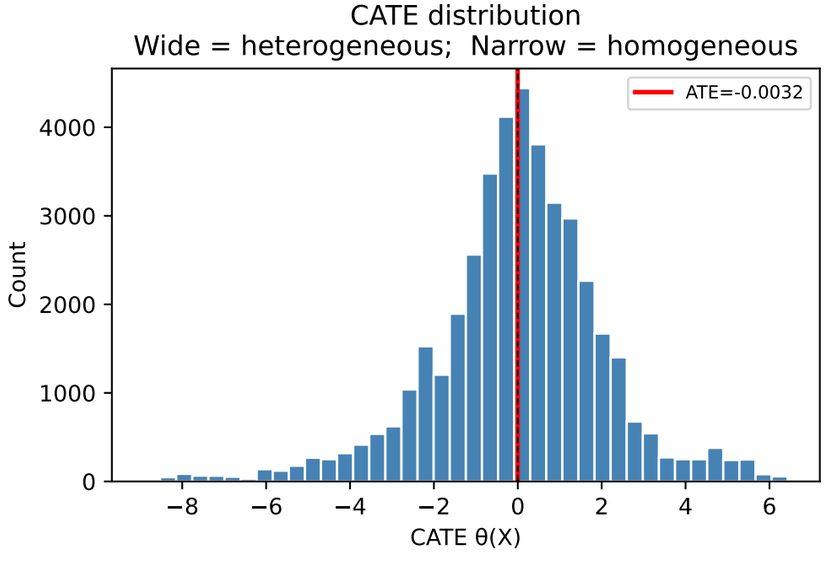


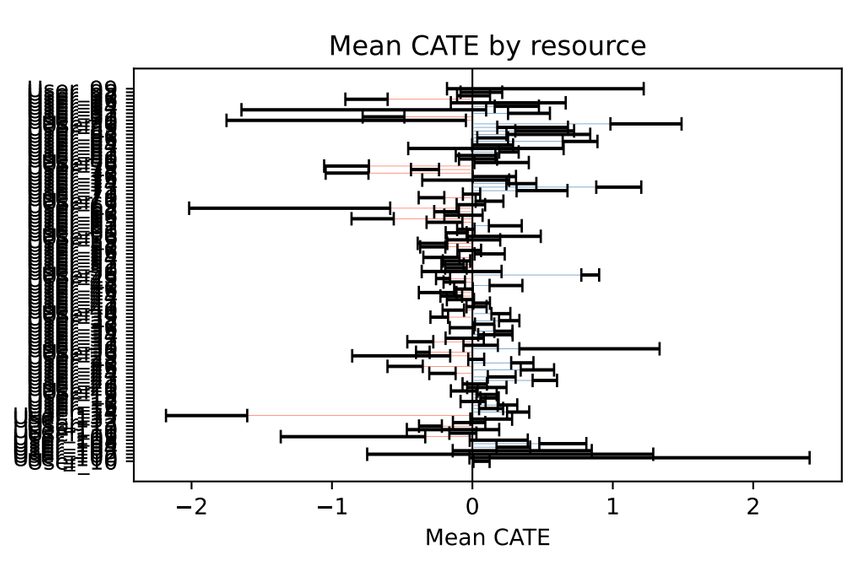

CATE — choice_set=(2, 3, 4) | transition='3' | n=41733 | ATE=0.2104 std=1.8642
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.2104
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

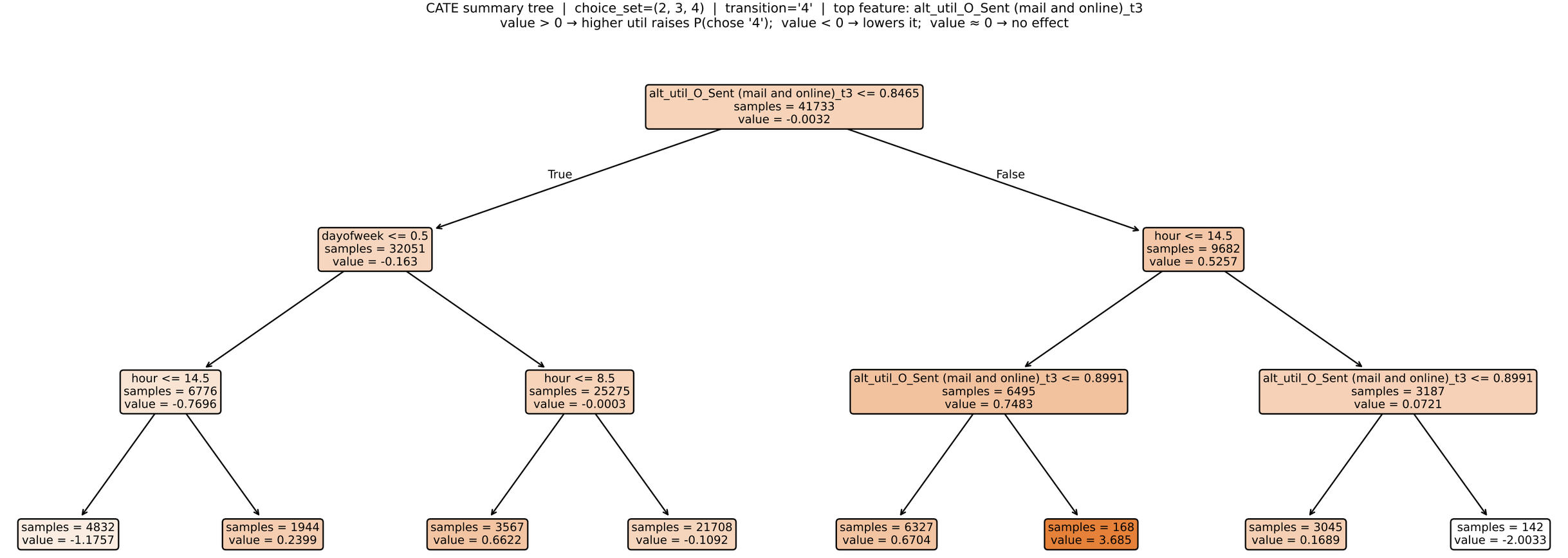

CATE summary tree | choice_set=(2, 3, 4) | transition='3' | top feature: alt_util_O_Sent (online only)_t4
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Sent (online only)_t4 <= 0.601
samples = 41733
value = 0.2104
True
False
dayofweek <= 2.5
samples = 1907
value = 1.8086
alt_util_O_Sent (online only)_t4 <= 0.6971
samples = 39826
value = 0.1338
alt_util_O_Sent (online only)_t4 <= 0.5337
samples = 1134
value = 2.3807
hour <= 10.5
samples = 773
value = 0.9693
alt_util_O_Sent (online only)_t4 <= 0.6394
samples = 6250
value = -0.4382
alt_util_O_Sent (online only)_t4 <= 0.7644
samples = 33576
value = 0.2403
samples = 341
value = 3.8453
samples = 793
value = 1.7509
samples = 216
value = 2.3938
samples = 557
value = 0.4169
samples = 2458
value = 0.4632
samples = 3792
value = -1.0224
samples = 7997
value = 0.6544
samples = 25579
value = 0.1109

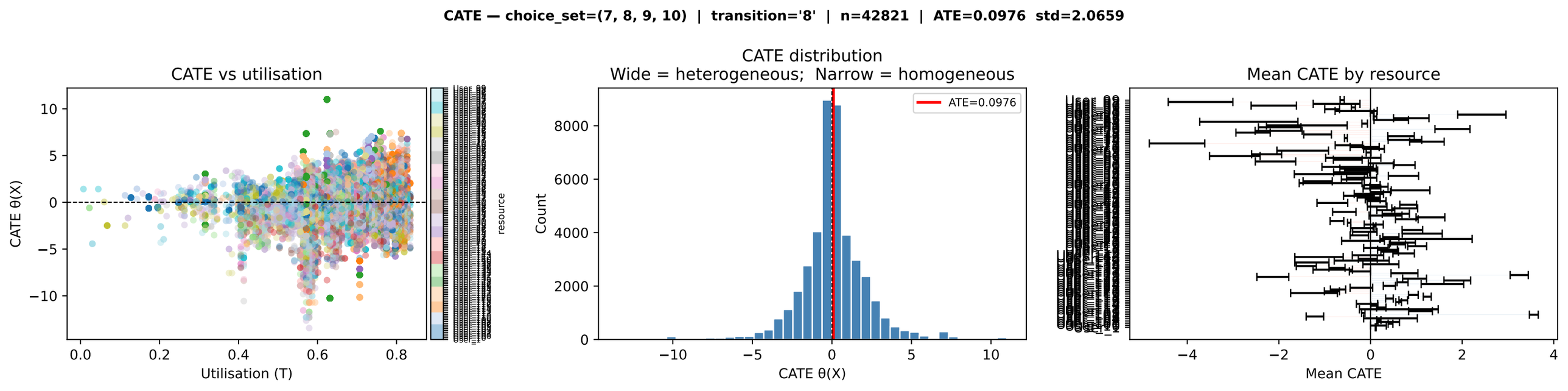
CATE — choice_set=(2, 3, 4) | transition='4' | n=41733 | ATE=-0.0032 std=2.0421

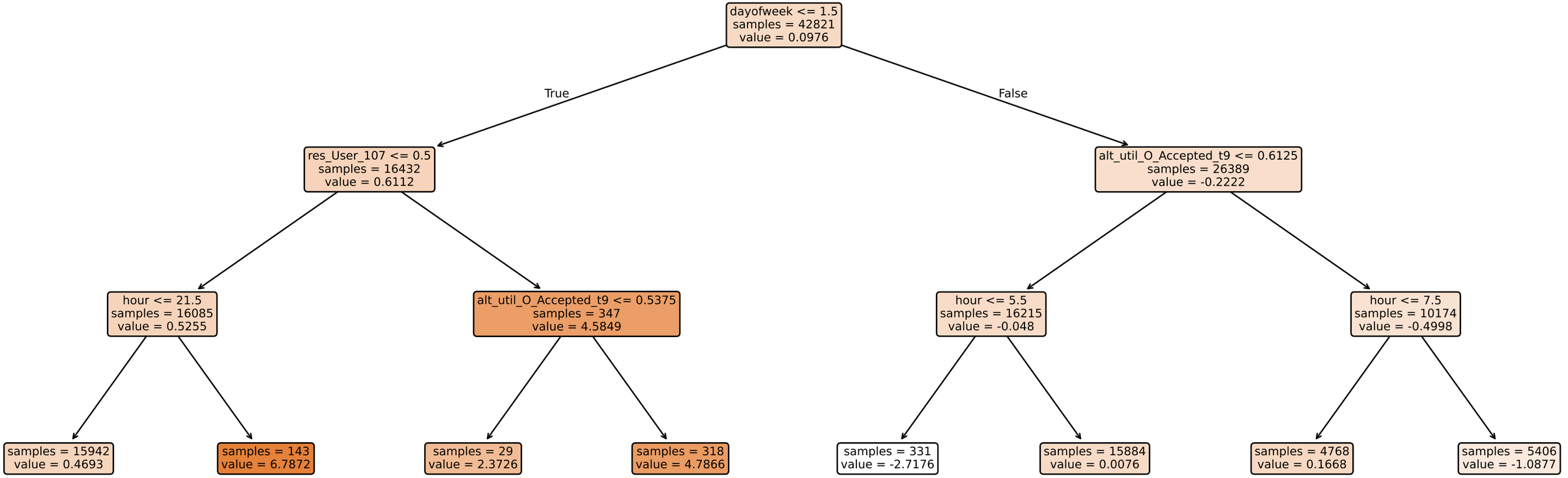


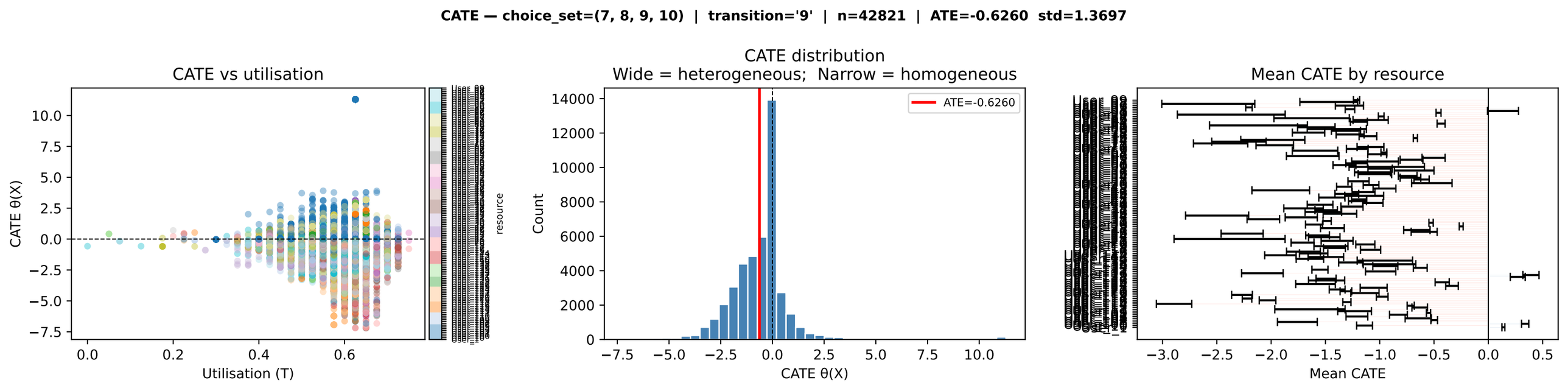


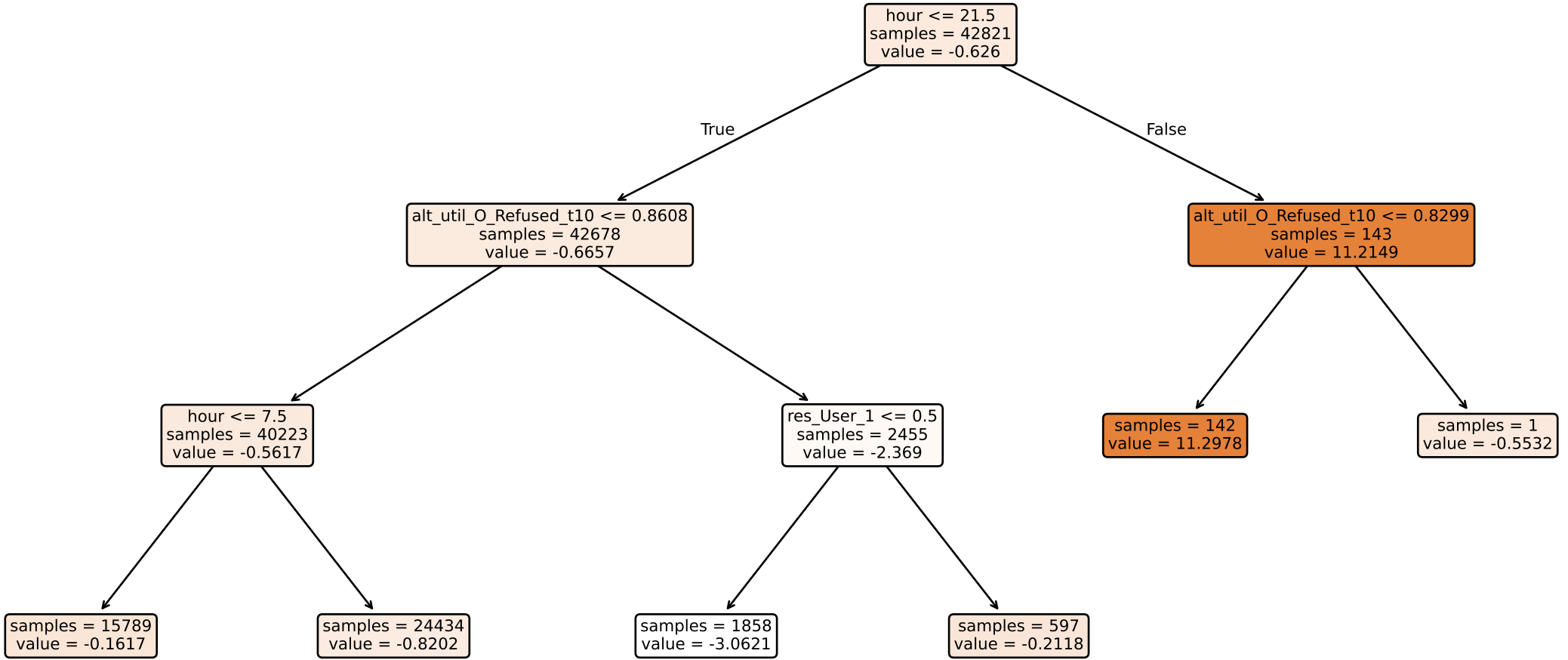


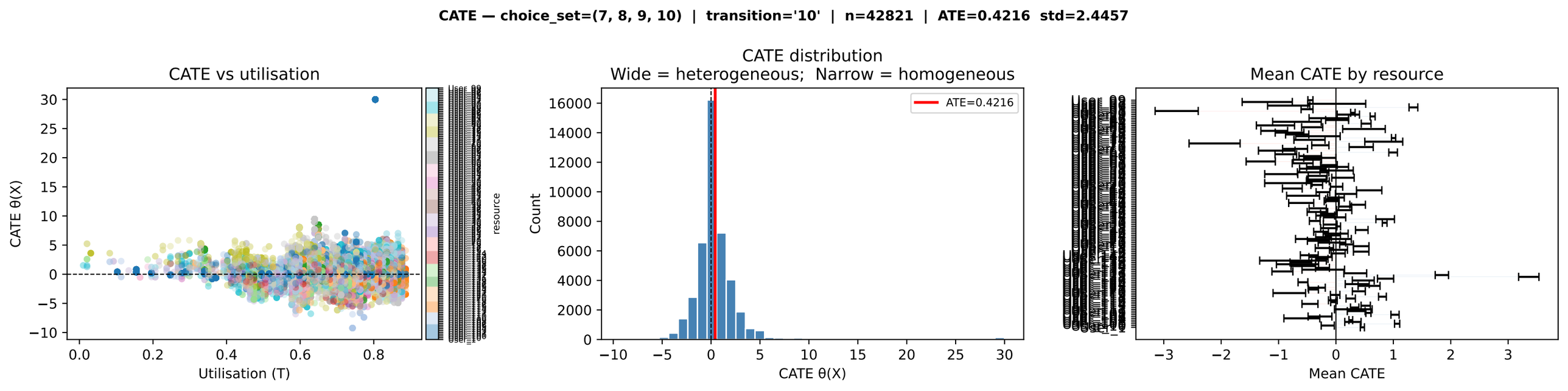

CATE summary tree | choice_set=(2, 3, 4) | transition='4' | top feature: alt_util_O_Sent (mail and online)_t3
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_O_Sent (mail and online)_t3 <= 0.8465
samples = 41733
value = -0.0032
True
False
dayofweek <= 0.5
samples = 32051
value = -0.163
hour <= 14.5
samples = 9682
value = 0.5257
hour <= 14.5
samples = 6776
value = -0.7696
hour <= 8.5
samples = 25275
value = -0.0003
alt_util_O_Sent (mail and online)_t3 <= 0.8991
samples = 6495
value = 0.7483
alt_util_O_Sent (mail and online)_t3 <= 0.8991
samples = 3187
value = 0.0721
samples = 4832
value = -1.1757
samples = 1944
value = 0.2399
samples = 3567
value = 0.6622
samples = 21708
value = -0.1092
samples = 6327
value = 0.6704
samples = 168
value = 3.685
samples = 3045
value = 0.1689
samples = 142
value = -2.0033

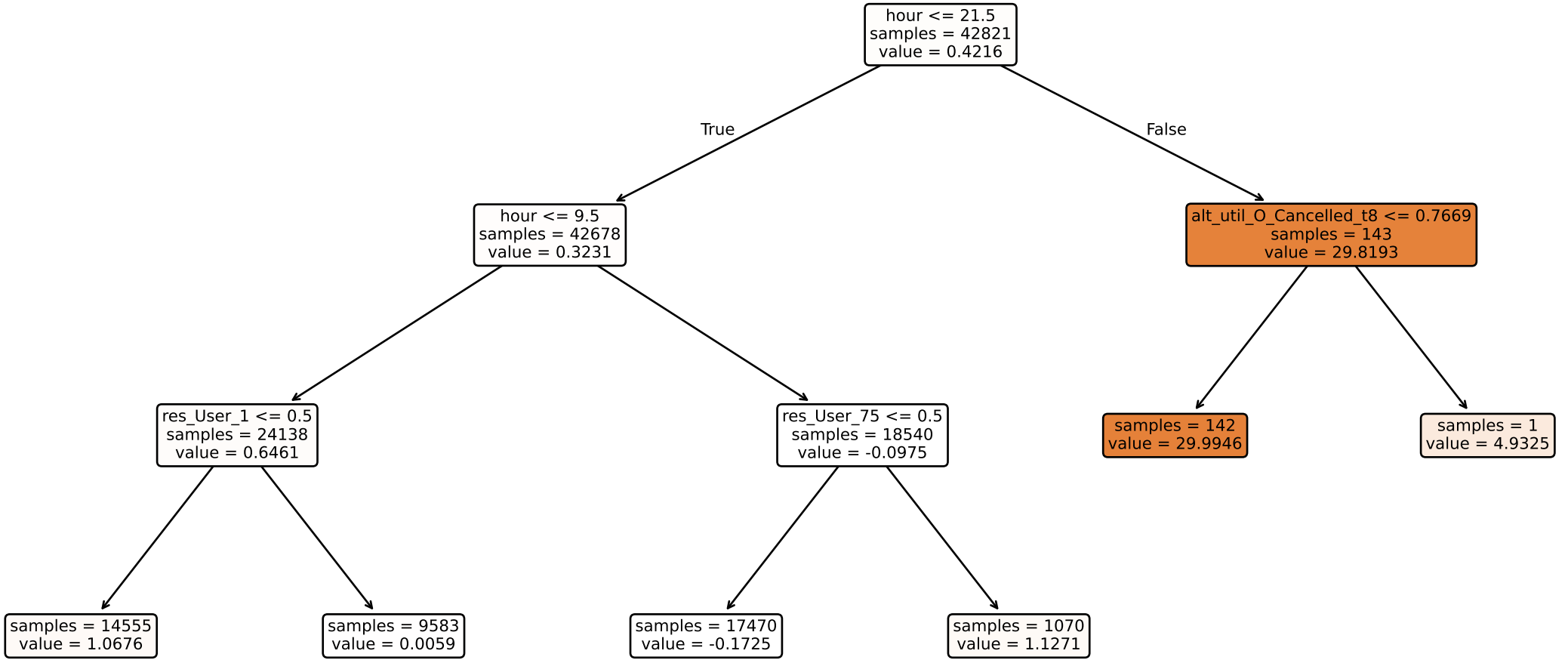

CATE — choice_set=(7, 8, 9, 10) | transition='8' | n=42821 | ATE=0.0976 std=2.0659
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.0976
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(7, 8, 9, 10) | transition='8' | top feature: hour
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 1.5
samples = 42821
value = 0.0976
True
False
res_User_107 <= 0.5
samples = 16432
value = 0.6112
alt_util_O_Accepted_t9 <= 0.6125
samples = 26389
value = -0.2222
hour <= 21.5
samples = 16085
value = 0.5255
alt_util_O_Accepted_t9 <= 0.5375
samples = 347
value = 4.5849
hour <= 5.5
samples = 16215
value = -0.048
hour <= 7.5
samples = 10174
value = -0.4998
samples = 15942
value = 0.4693
samples = 143
value = 6.7872
samples = 29
value = 2.3726
samples = 318
value = 4.7866
samples = 331
value = -2.7176
samples = 15884
value = 0.0076
samples = 4768
value = 0.1668
samples = 5406
value = -1.0877


CATE — choice_set=(7, 8, 9, 10) | transition='9' | n=42821 | ATE=-0.6260 std=1.3697
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.6260
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(7, 8, 9, 10) | transition='9' | top feature: hour
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 21.5
samples = 42821
value = -0.626
True
False
alt_util_O_Refused_t10 <= 0.8608
samples = 42678
value = -0.6657
alt_util_O_Refused_t10 <= 0.8299
samples = 143
value = 11.2149
hour <= 7.5
samples = 40223
value = -0.5617
res_User_1 <= 0.5
samples = 2455
value = -2.369
samples = 142
value = 11.2978
samples = 1
value = -0.5532
samples = 15789
value = -0.1617
samples = 24434
value = -0.8202
samples = 1858
value = -3.0621
samples = 597
value = -0.2118


CATE — choice_set=(7, 8, 9, 10) | transition='10' | n=42821 | ATE=0.4216 std=2.4457
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.4216
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(7, 8, 9, 10) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 21.5
samples = 42821
value = 0.4216
True
False
hour <= 9.5
samples = 42678
value = 0.3231
alt_util_O_Cancelled_t8 <= 0.7669
samples = 143
value = 29.8193
res_User_1 <= 0.5
samples = 24138
value = 0.6461
res_User_75 <= 0.5
samples = 18540
value = -0.0975
samples = 142
value = 29.9946
samples = 1
value = 4.9325
samples = 14555
value = 1.0676
samples = 9583
value = 0.0059
samples = 17470
value = -0.1725
samples = 1070
value = 1.1271

| | | |

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(2, 3, 4)	3	41733	0.210368	1.86423	-11.37915	-0.817347	0.214258	1.283539	8.165874	56.2	43.8	alt_util_O_Sent (online only)_t4
(2, 3, 4)	4	41733	-0.003188	2.042086	-8.910895	-0.998913	0.081568	1.15095	6.433925	52.2	47.8	alt_util_O_Sent (mail and online)_t3
(7, 8, 9, 10)	8	42821	0.097606	2.065858	-13.45143	-0.755755	0.011003	0.964959	10.996384	50.9	49.1	hour
(7, 8, 9, 10)	9	42821	-0.625986	1.369677	-7.200097	-1.337844	-0.384453	0.002059	11.297751	26.6	73.4	hour
(7, 8, 9, 10)	10	42821	0.421562	2.445666	-9.237517	-0.510017	0.071117	1.122895	29.99458	56.5	43.5	hour

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md